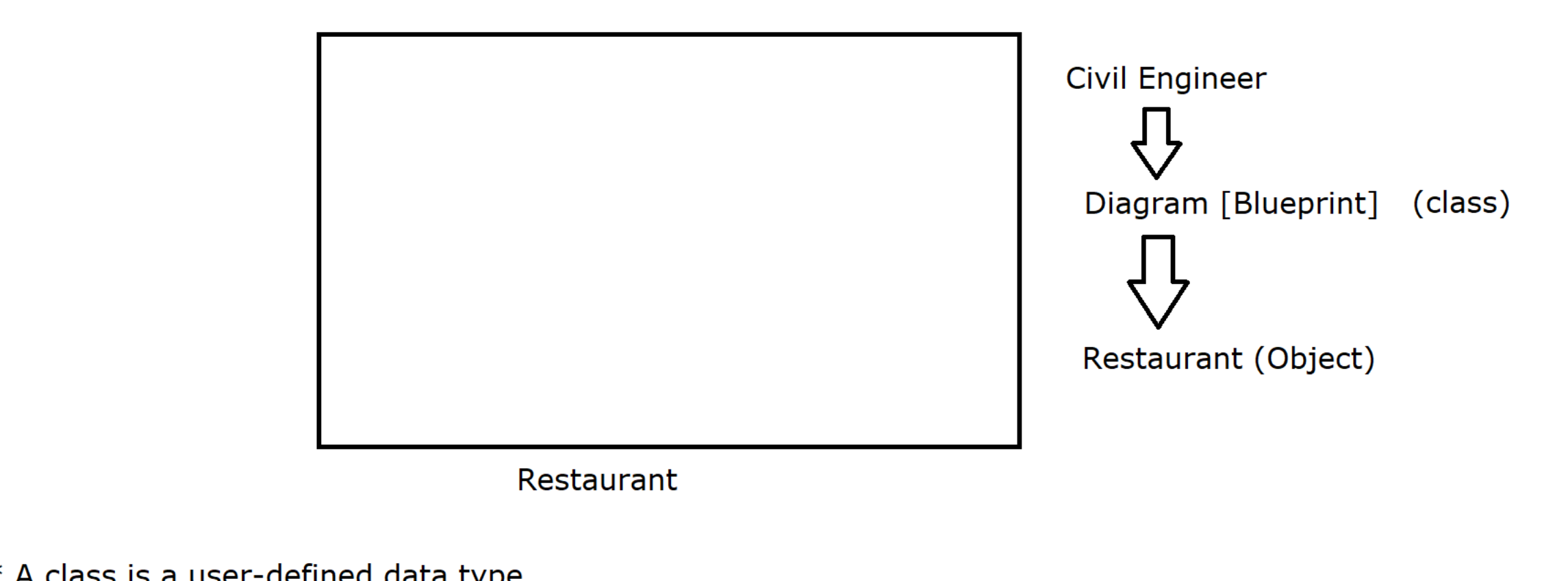


What is a class ?

* A class is a blueprint OR template OR prototype for creating an Object.

Example :



* A class is a user-defined data type.

* A class is a component which is used to define the OBJECT PROPERTIES + OBJECT BEHAVIOUR

Example:

```
public class Student
{
    Object Properties [Student properties -> Non staic Field]

    +

    Object Behaviour [Student functionality -> Non static method]
}
```

Steel Organization

Hammer

Blacksmith

1 HR : To gather the raw material (Iron raw)

2 HRS : To boil the Iron raw (into liquid)

4 HRS : To beat & provide Hammer shape

7 HRS : Hammer

100 X 7 = 700 HRS

Engineer

1 HR : To gather the raw material (Iron raw)

2 HRS : To boil the Iron raw (into liquid)

1 HR : To create a dye diagram. class

3 HRS : To create original dye. object

1 HR : To cast the liquid (Iron raw)

8 HRS : Hammer

100 X 4 = 400 HRS

WAP to initialize the non static field (vizag) wit the help of reference variable :

[Single file approach]

```
package com.ravi.oop;

//Single file approach
//ELC [1st class]
public class StudentDemo
{
    void main()
    {
        Student raj = new Student();
        //Initializing the object properties using reference variable
        raj.studentName = "Raj Gourav";
        raj.rollNumber = 101;
        raj.studentHeight = 5.8;

        //Calling the behavior
        raj.talk();
        raj.writeExam();

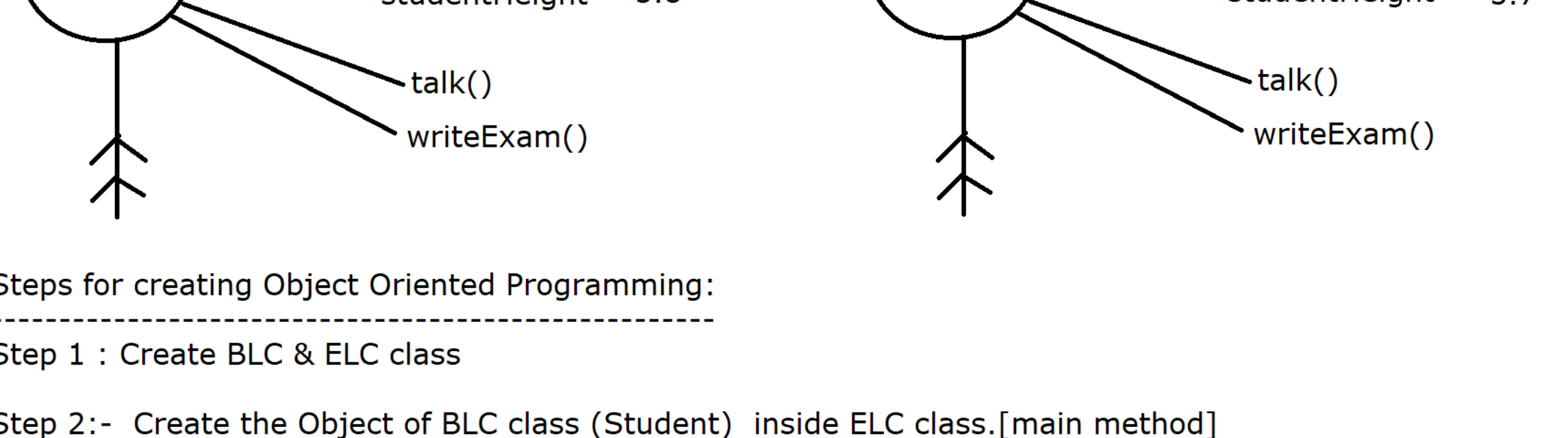
        Student priya = new Student();
        priya.studentName = "Priya";
        priya.studentHeight = 5.7;
        priya.rollNumber = 102;

        priya.talk();
        priya.writeExam();
    }
}

//BLC
class Student
{
    //Object properties (Non static fields)
    String studentName;
    int rollNumber;
    double studentHeight;

    //Object behavior [Non static method]
    public void talk()
    {
        IO.println("Hi everyone, My name is :"+studentName);
        IO.println("My roll number is :"+rollNumber);
        IO.println("My height is :"+studentHeight);
    }

    public void writeExam()
    {
        IO.println("My name is :"+studentName+", Every saturday I write Exam ");
        IO.println("After exam, I enjoy my weekend");
    }
}
```



- Steps for creating Object Oriented Programming:
- Step 1 : Create BLC & ELC class
 - Step 2:- Create the Object of BLC class (Student) inside ELC class.[main method]
 - Step 3 :- Define all the object properties and behaviour inside the BLC class based on your imagination. [Thinking]
 - Step 4 :- Initialize all the object properties with user friendly value by using reference variable. [raj.rollNumber = 111]
 - step 5 :- call the behaviour (calling the methods raj.talk())

Non static Field OR Instance Variable :

- * It is a class level variable.
 - * If a non static variable (not declared with static modifier) is defined **inside a class but outside of the method** then **It is called non static field.**
- Example :
- ```
public class Student
{
 //Non static fields
 int rollNumber;
 String studentName;
 double height;
}
```
- \* As far as its accessibility is concerned, A non static field we can access any-where from the class as well as **outside of the class also if not declared private.** [If private then within the class only]
  - \* The life of a non static field will be started during the object creation (at the time of object creation) that means we cannot think about non static field without object creation. [If no object, no non static field]

How to initialize the non static field (vizag) using method with parameter :

```
package com.ravi.oop;

//ELC
public class EmployeeDemo
{
 public static void main(String[] args)
 {
 Employee scott = new Employee();
 scott.setEmployeeData(101, "Mr Scott", 90000);
 scott.showEmployeeData();
 }
}

//BLC
class Employee
{
 int employeeId;
 String employeeName;
 double employeeSalary;

 public void setEmployeeData(int id, String name, double salary)
 {
 employeeId = id;
 employeeName = name;
 employeeSalary = salary;
 }

 public void showEmployeeData()
 {
 IO.println("Employee Id is :"+employeeId);
 IO.println("Employee Name is :"+employeeName);
 IO.println("Employee Salary is :"+employeeSalary);
 }
}
```